A SHEAF OF SETS INJECTS INTO THE PRODUCT OF ITS STALKS OVER AN OPEN SET

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Sheaf condition on X: two sections agreeing on every member of an open covering are equal. invoked at proof lines 1179-1181. Provides the uniqueness step concluding s = s' from the covering by the neighbourhoods V‸x.

Lemma 0.2. *Let $\mathcal{F}$ be a sheaf of sets on the topological space X. For every open $U \subset X$ the map*

$$\mathcal{F}(U) \longrightarrow \prod\nolimits_{x \in U} \mathcal{F}_x$$

is injective.

Proof. Suppose that $s, s' \in \mathcal{F}(U)$ map to the same element in every stalk $\mathcal{F}_x$ for all $x \in U$. This means that for every $x \in U$, there exists an open $V^x \subset U$, $x \in V^x$ such that $s|_{V^x} = s'|_{V^x}$. But then $U = \bigcup_{x \in U} V^x$ is an open covering. Thus by the uniqueness in the sheaf condition we see that $s = s'$. □

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